

TKX-50

TKX-50 is the chemical compound with the formula $\text{C}_2\text{H}_8\text{N}_{10}\text{O}_4$; it is a hydroxylammonium salt of 5,5'-bistetrazolyl-1,1'-diolate that forms colorless crystals, and it is a high-performing explosive with superior performance to RDX and HMX while being more resistant to mechanical stimuli. It is also considered an environmentally-friendly replacement for several conventional explosives.^{[1][2]} As of 2014, the US Army was preparing it in 20 kg batches in order to evaluate it as an explosive.^[3] It was invented by Niko Fischer, Dennis Fischer, Davin G. Piercey, Thomas M. Klapötke and Jörg Stierstorfer in Klapötke's research group at LMU Munich.^[4] Its development was funded by LMU Munich, the U.S. Army Research Laboratory (ARL), the Armament Research, Development and Engineering Center (ARDEC), the Strategic Environmental Research and Development Program (SERDP) and the Office of Naval Research (ONR).

Synthesis

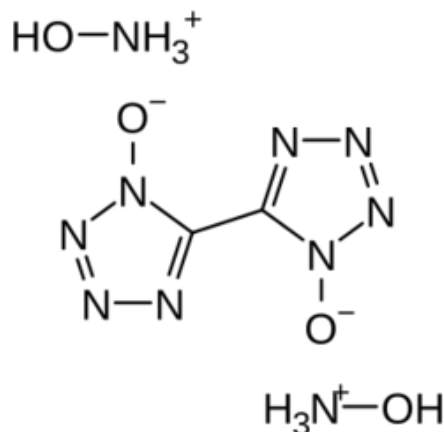
The compound is prepared by reacting 5,5'-bistetrazolyl-1,1'-diol (BTO) with hydroxylamine. The starting compound BTO can be obtained relatively easily from glyoxal, hydroxylamine, chlorine, and sodium azide.^{[5] [6]}

Since the initial synthesis of TKX-50, multiple research groups have improved the synthesis to make it more suitable for mass production, including by adding protecting groups to make intermediates safer,^[7] and the United States Army Research Laboratory has performed multiple steps of the synthesis in a 1-pot reaction.^[8]

Characteristics

TKX-50 is a crystalline solid with no melting point. The compound occurs in two polymorphic crystal forms. At about 180 °C, a second-order phase

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Names

IUPAC name

Dihydroxylammonium-5,5'-bistetrazolyl-1,1'-diolate

Other names

Dihydroxylammonium-5,5'-bistetrazolyl-1,1'-diolate, Dihydroxylammonium-5,5'-bistetrazolyl-1,1'-dioxide,

Identifiers

CAS Number

1403467-86-6 (https://comchem.org/detail?cas_rn=1403467-86-6) ✓

3D model (JSmol)

Interactive image (<https://chemapps.stolaf.edu/jmol/jmol.php?model=%5BNH3%2B%5DO.%5BNH3%2B%5DO.%5BO-%5DN1N%3DNN%3DC1C2%3DNN%3DNN2%5BO-%5D>)

PubChem CID

139051543 (<https://pubchem.ncbi.nlm.nih.gov/compound/139051543>)

InChI

InChI=1S/C2N8O2.2H4NO/c11-9-1(3-5-7-9)2-4-6-8-10(2)12;2*1-2/h;2*2H,1H3/q-2;2*+1
Key: UHTGTYVZCBKFMU-UHFFFAOYSA-N

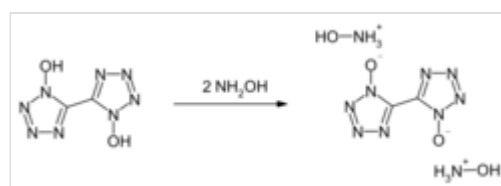
SMILES

transition with a small heat capacity change of about $0.3 \text{ J Mol}^{-1}\cdot\text{K}^{-1}$ is observed, resulting in a transformation from the low-temperature to a high-temperature crystal form. The crystal system changes from monoclinic with the space group $P 2_1/c$ to triclinic with the space group $P 1$.^[9] The compound is thermally unstable. DSC measurements show an exothermic decomposition reaction above $215 \text{ }^\circ\text{C}$ with a latent heat of decomposition of -2200 kJ kg^{-1} and -520 kJ mol^{-1} , respectively.^{[10][11]} The standard enthalpy of formation based on combustion calorimetry measurements is 473 kJ mol^{-1} .^[12]

Explosive uses

TKX-50 has been found to lead to melt-cast formulations with higher performance than those with HMX.^[13] Its detonation velocity, Chapman–Jouguet detonation pressure, and density are also superior to RDX.^{[1][2]}

[NH3+] ₂ O.[NH3+] ₂ O.[O-] ₂ N1N=NN=C1C2=NN=NN2 [O-]	
Properties	
Chemical formula	C ₂ H ₈ N ₁₀ O ₄
Molar mass	236.15 g/mol
Appearance	colorless crystals
Density	1.92 g/cm ³
Explosive data	
Shock sensitivity	20 J
Friction sensitivity	120 N
Detonation velocity	9698 m/s
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at $25 \text{ }^\circ\text{C}$ [$77 \text{ }^\circ\text{F}$], 100 kPa).	
Infobox references	



TKX-50 synthesis

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